1. INTRODUCTION

Aquatic life refers to the diverse range of organisms that live in water environments such as oceans, rivers, lakes, and ponds. These ecosystems support a wide variety of living beings, including fish, aquatic plants, microorganisms, and invertebrates, all of which interact in a delicate ecological balance. Water serves as a fundamental medium that influences the biological, chemical, and physical processes essential for sustaining life. The health and productivity of aquatic ecosystems are directly linked to environmental parameters such as temperature, dissolved oxygen, turbidity, pH, and nutrient availability.

Fish are among the most significant components of aquatic life, playing a vital role in both ecological balance and human food supply. Aquaculture, the practice of farming fish and other aquatic organisms, has become an essential sector to meet the growing global demand for protein-rich food. However, maintaining optimal environmental conditions in aquaculture systems is critical, as even slight variations in water quality can adversely affect fish health, growth, and survival. Factors such as temperature fluctuations, poor water clarity, and inadequate oxygen levels can lead to stress, disease, and reduced productivity.

Traditionally, monitoring of aquatic environments has been carried out manually using basic tools and periodic observations. Such methods are often inefficient, time-consuming, and prone to human error. Moreover, they fail to provide continuous insights into dynamic environmental changes, which are crucial for timely decision-making. As a result, there is an increasing need for automated and intelligent monitoring systems that can ensure sustainable management of aquatic ecosystems.

With advancements in technology, modern approaches now integrate sensors, embedded systems, and data communication techniques to enable real-time monitoring of water parameters. These systems can continuously collect and analyze data, providing accurate and up-to-date information about the aquatic environment. Additionally, the integration of imaging technologies and computer vision allows for the observation and analysis of fish behavior, offering deeper insights into their activity patterns and overall health.

In recent years, the combination of Internet of Things (IoT) and embedded vision has revolutionized the field of aquaculture monitoring. By deploying smart devices capable of sensing, processing, and transmitting data, it is possible to create efficient, cost-effective, and scalable solutions for managing aquatic life. Such systems not only reduce manual effort but also enhance precision and reliability, ultimately contributing to improved fish yield and environmental sustainability.

1.1 Introduction to Aquaculture

Aquaculture, also known as fish farming, is the practice of cultivating aquatic organisms such as fish, crustaceans, mollusks, and aquatic plants under controlled environmental conditions. It has emerged as one of the fastest-growing sectors in global food production, playing a crucial role in meeting the increasing demand for seafood due to population growth and the decline of natural fish stocks. Aquaculture contributes significantly to food security, nutrition, and economic development, especially in developing countries where fish is a major source of protein.

The success of aquaculture largely depends on maintaining optimal water quality and environmental conditions. Parameters such as water temperature, oxygen levels, turbidity, pH, and water depth directly influence the health, growth rate, and survival of aquatic organisms. Even minor deviations from ideal conditions can lead to stress, disease outbreaks, and reduced productivity. Therefore, continuous monitoring and proper management of these parameters are essential for sustainable aquaculture practices.

Traditionally, aquaculture management relies on manual observation and periodic testing of water conditions. Farmers often use basic instruments or visual inspection to assess water quality and fish behavior. However, these conventional methods are time-consuming, labor-intensive, and lack real-time responsiveness. As a result, critical changes in the environment may go unnoticed, leading to significant losses in fish yield and overall system efficiency.

With the advancement of technology, modern aquaculture systems are increasingly adopting automated monitoring solutions. These systems utilize sensors, embedded controllers, and communication technologies to continuously measure environmental parameters and provide real-time data. The integration of Internet of Things (IoT) has enabled remote monitoring and control, allowing farmers to make informed decisions quickly and efficiently. Additionally, the use of cameras and computer vision techniques facilitates the observation of fish behavior, feeding patterns, and health conditions without disturbing the aquatic environment.

Smart aquaculture systems not only improve productivity but also promote sustainable resource management by reducing waste, optimizing feed usage, and minimizing environmental impact. By leveraging technology, aquaculture can become more efficient, reliable, and scalable, addressing the challenges of traditional farming methods. This project aligns with these advancements by proposing an intelligent system that integrates sensor-based monitoring and visual analysis to enhance aquaculture management.

1.2 Difficulties in Aquaculture

Aquaculture, while being a rapidly growing and essential sector for global food production, faces several challenges that affect its efficiency, sustainability, and profitability. One of the primary difficulties is maintaining optimal water quality, which is critical for the health and growth of aquatic organisms. Parameters such as temperature, dissolved oxygen, turbidity, pH, and ammonia levels must remain within specific ranges. However, these parameters are highly dynamic and can fluctuate due to environmental conditions, overfeeding, or waste accumulation. Poor water quality can lead to stress, disease outbreaks, and even mass mortality of fish.

Another significant challenge is the lack of real-time monitoring systems in traditional aquaculture practices. Most small and medium-scale fish farms rely on manual inspection and periodic testing, which are time-consuming and prone to human error. This delay in detecting unfavorable conditions often results in late corrective actions, leading to reduced productivity and financial losses. Continuous monitoring is essential but not always feasible with conventional methods.

Fish health management is also a major concern in aquaculture. Diseases can spread rapidly in densely populated ponds or tanks, causing severe economic damage. Early detection of abnormal fish behavior, such as reduced movement or irregular swimming patterns, is difficult through manual observation alone. Moreover, environmental stress caused by poor water conditions further weakens fish immunity, making them more susceptible to infections.

Resource management, particularly feeding practices, presents another difficulty. Overfeeding not only increases operational costs but also contributes to water pollution through uneaten feed and waste. On the other hand, underfeeding can hinder fish growth and reduce yield. Achieving the right balance requires accurate monitoring and understanding of fish behavior and environmental conditions.

Energy consumption and infrastructure limitations also pose challenges, especially in remote or rural aquaculture setups. Continuous operation of aerators, pumps, and monitoring equipment requires a stable power supply, which may not always be available. Additionally, limited access to advanced technologies and high initial setup costs can prevent small-scale farmers from adopting modern aquaculture solutions.

Environmental factors such as climate change, water scarcity, and pollution further complicate aquaculture operations. Changes in weather patterns can affect water temperature and quality, while contamination from external sources can degrade the aquatic environment. These external influences are often unpredictable and difficult to control.

In summary, aquaculture faces multiple challenges, including water quality management, lack of real-time monitoring, disease control, inefficient resource utilization, and environmental uncertainties. Addressing these difficulties requires the integration of advanced technologies such as sensors, automation, and intelligent monitoring systems. The development of smart aquaculture solutions can help overcome these challenges by providing accurate, real-time data and enabling proactive decision-making.

1.3 Introduction to Deep Sea Fishing

Deep sea fishing refers to the practice of fishing in the deeper parts of oceans, typically beyond the continental shelf, where water depths exceed several hundred meters. Unlike coastal or inland fishing, deep sea fishing targets species that inhabit the open ocean and deeper waters, such as tuna, swordfish, cod, and various shellfish. This method of fishing plays a significant role in global seafood supply and contributes substantially to the economy of many coastal nations.

The deep sea environment is vastly different from shallow water ecosystems. It is characterized by high pressure, low temperatures, minimal light penetration, and limited nutrient availability. These extreme conditions make deep sea fishing a complex and challenging activity that requires specialized equipment, vessels, and advanced navigation systems. Modern deep sea fishing operations rely on technologies such as sonar, GPS, and automated fishing gear to locate and harvest fish efficiently.

One of the major challenges in deep sea fishing is the difficulty in locating fish populations due to the vastness and depth of the ocean. Fish behavior in deep waters is less predictable, and environmental conditions can change rapidly. Additionally, the absence of natural light makes visual observation nearly impossible, requiring the use of artificial lighting and imaging technologies. The harsh conditions also pose risks to equipment durability and human safety.

Sustainability is another critical concern in deep sea fishing. Overfishing and destructive fishing practices can severely impact marine ecosystems, leading to the depletion of fish stocks and disruption of ecological balance. Deep sea species often have slower growth rates and longer lifespans, making them more vulnerable to overexploitation. Therefore, there is an increasing emphasis on adopting sustainable fishing practices and monitoring techniques to ensure long-term resource availability.

In recent years, technological advancements have played a vital role in improving deep sea fishing operations. The integration of sensors, remote monitoring systems, and data analytics has enabled better tracking of environmental conditions and fish movement. These innovations help optimize fishing efforts, reduce operational costs, and minimize environmental impact. Furthermore, the use of imaging systems and computer vision techniques is being explored to study underwater life and enhance fish detection capabilities.

1.4 Bottom Trawling

Destructive fishing methods have become a major concern in modern marine ecosystems due to their severe impact on biodiversity and environmental sustainability. Among these methods, bottom trawling—often informally referred to as “double net fishing”—is one of the most harmful practices. It involves dragging large, weighted fishing nets across the seabed to capture fish and other marine organisms. These nets are designed to cover wide areas and collect everything in their path, regardless of species or size.

Bottom trawling is widely used in commercial fishing because of its high efficiency and large catch volume. However, this efficiency comes at a significant environmental cost. As the nets move along the ocean floor, they disturb and destroy critical habitats such as coral reefs, seagrass beds, and breeding grounds of marine life. These ecosystems play a vital role in maintaining marine biodiversity and supporting fish populations. Their destruction leads to long-term ecological damage, which may take decades to recover.

One of the major issues associated with bottom trawling is bycatch, which refers to the unintended capture of non-target species. Along with commercially valuable fish, the nets often trap juvenile fish, endangered species, and other marine organisms such as turtles, crustaceans, and plants. A large portion of this bycatch is discarded, often dead or severely injured, resulting in significant waste and loss of marine life. This not only reduces biodiversity but also affects the natural balance of the ecosystem.

Furthermore, bottom trawling contributes to the depletion of fish stocks by capturing immature fish before they can reproduce. This disrupts the natural life cycle and leads to a decline in fish populations over time. The practice also stirs up sediments from the seabed, increasing water turbidity and releasing trapped carbon, which can further impact marine environments and contribute to climate-related concerns.

In recent years, there has been growing awareness about the negative impacts of destructive fishing practices. Environmental organizations and regulatory bodies are promoting sustainable alternatives and stricter regulations to protect marine ecosystems. The adoption of smart monitoring systems, improved fishing techniques, and aquaculture-based solutions can help reduce dependency on such harmful methods.

1.5 Effects and Prevention of Destructive Fishing Methods

Bottom trawling is considered one of the most destructive fishing practices due to its widespread environmental, ecological, and economic impacts. While it is efficient in terms of catch volume, the long-term consequences significantly outweigh its short-term benefits. Understanding these effects is essential for developing sustainable alternatives and effective prevention strategies.

1.5.1 Effects of Bottom Trawling

One of the most severe effects of bottom trawling is the destruction of marine habitats. As heavy nets are dragged across the ocean floor, they physically damage sensitive ecosystems such as coral reefs, sponges, and seagrass beds. These habitats serve as breeding and feeding grounds for many marine species, and their destruction leads to a decline in biodiversity. Since many deep-sea ecosystems grow very slowly, recovery can take several decades or may not occur at all.

Another major impact is the issue of bycatch. Bottom trawling is a non-selective fishing method, meaning it captures not only the target species but also a large number of unintended organisms. These include juvenile fish, endangered species, and other marine life such as turtles and crustaceans. A significant portion of this bycatch is discarded, often dead, resulting in wasteful depletion of marine resources. This reduces fish populations and disrupts the ecological balance.

Bottom trawling also contributes to overfishing by removing large quantities of fish, including immature individuals that have not yet reproduced. This leads to a rapid decline in fish stocks and threatens the sustainability of fisheries. Over time, this can result in economic losses for fishing communities due to reduced yields.

Additionally, the process of dragging nets across the seabed disturbs sediments, increasing water turbidity and releasing stored carbon into the water column. This can negatively affect water quality and contribute to climate change by increasing carbon emissions. The disturbance also impacts organisms living within the sediment, further affecting the marine food chain.

1.5.2 Prevention and Sustainable Solutions

To mitigate the harmful effects of bottom trawling, several preventive measures and sustainable alternatives have been proposed. One of the most effective approaches is the implementation of strict regulations and fishing policies. Governments and international organizations can enforce bans or restrictions on bottom trawling in ecologically sensitive areas such as coral reefs and marine protected zones.

The adoption of selective fishing techniques is another important solution. Methods such as hook-and-line fishing, traps, and modified nets can reduce bycatch and minimize damage to the seabed. These techniques allow fishermen to target specific species while preserving the surrounding ecosystem.

Technological advancements also play a crucial role in prevention. The use of sonar, GPS, and sensor-based monitoring systems enables more precise identification of fish populations, reducing the need for indiscriminate fishing methods. Additionally, integrating computer vision and underwater imaging can help monitor fish behavior and improve targeted fishing practices.

Aquaculture presents a sustainable alternative to traditional fishing. By cultivating fish in controlled environments, it reduces the pressure on natural fish stocks and minimizes the need for destructive fishing methods. Smart aquaculture systems, such as those using IoT and embedded monitoring technologies, further enhance efficiency and environmental sustainability.

Public awareness and education are equally important in promoting responsible fishing practices. Encouraging consumers to choose sustainably sourced seafood can drive demand for environmentally friendly methods and discourage harmful practices like bottom trawling.

1.6 Difficulties in Pond Fishing

Pond fishing, a common method in small-scale aquaculture, involves cultivating fish in controlled water bodies such as ponds or tanks. While it is relatively simple and cost-effective compared to large-scale aquaculture or deep-sea fishing, pond fishing faces several challenges that affect productivity, fish health, and overall system efficiency.

One of the primary difficulties in pond fishing is maintaining water quality. Parameters such as temperature, dissolved oxygen, turbidity, pH, and ammonia levels must be kept within optimal ranges for fish survival and growth. However, these factors are highly sensitive to environmental changes such as weather variations, rainfall, and evaporation. Poor water quality can lead to stress, disease, and even mass mortality of fish, making regular monitoring essential.

Another significant challenge is the lack of continuous monitoring. In many traditional pond systems, farmers rely on manual inspection and periodic testing to assess water conditions and fish health. This approach is time-consuming and often fails to detect sudden changes in the environment. As a result, critical issues such as oxygen depletion or water contamination may go unnoticed until they cause serious damage.

Fish health management is also a major concern. In pond environments, fish are often densely populated, which increases the risk of disease transmission. Identifying early signs of illness or abnormal behavior is difficult without proper monitoring tools. Delayed detection can lead to rapid spread of diseases, resulting in economic losses.

Feeding management presents another difficulty in pond fishing. Providing the right amount of feed is crucial for optimal growth. Overfeeding can lead to water pollution due to leftover feed and increased waste, while underfeeding can stunt fish growth. Achieving the correct balance requires careful observation and experience, which may not always be accurate.

Additionally, environmental factors such as algae growth and sediment accumulation can negatively impact pond conditions. Excessive algae can reduce oxygen levels, especially during nighttime, leading to fish stress. Sediment buildup can affect water clarity and harbor harmful microorganisms.

Infrastructure and resource limitations also pose challenges. Many pond fishing setups lack advanced equipment for aeration, filtration, and monitoring. Power supply issues and limited access to technology further restrict the adoption of modern solutions, especially in rural areas.

1.7 Water Quality in Pond Ecosystems

Water quality in pond ecosystems is a critical factor that directly influences the health, growth, and survival of fish and other aquatic organisms. Ponds are relatively small and enclosed water bodies, making them highly sensitive to environmental changes and internal disturbances. Key parameters such as temperature, dissolved oxygen (DO), pH, turbidity, and nutrient levels must be carefully maintained within optimal ranges.

Temperature plays a vital role in regulating metabolic activities of fish. In pond environments, temperature can fluctuate significantly due to weather conditions, affecting fish behavior and oxygen solubility. Dissolved oxygen is another crucial parameter, as fish rely on it for respiration. Low oxygen levels, especially during night or due to excessive algae growth, can lead to stress or suffocation.

Turbidity, caused by suspended particles or algae, affects light penetration and photosynthesis in the pond. High turbidity levels can reduce oxygen production and hinder fish feeding. Additionally, nutrient accumulation from overfeeding or waste can lead to eutrophication, resulting in excessive algae growth and degraded water quality.

Since ponds lack natural water circulation, pollutants and waste materials tend to accumulate over time. This makes regular monitoring and management essential to maintain a stable aquatic environment. Proper aeration, controlled feeding, and periodic water exchange are commonly used practices to improve pond water quality.

1.8 Water Quality in Marine (Sea) Environments

Marine water quality is influenced by large-scale environmental factors and plays a vital role in sustaining ocean ecosystems. Unlike ponds, the sea has vast water volume and natural circulation systems such as tides and currents, which help maintain relatively stable conditions. However, marine ecosystems are still vulnerable to pollution, climate change, and human activities.

Salinity is a defining parameter in marine water, typically ranging between 30 to 35 parts per thousand (ppt). Marine organisms are adapted to specific salinity levels, and sudden changes can cause stress or mortality. Temperature in ocean environments varies with depth and location, affecting species distribution and metabolic rates.

Dissolved oxygen levels in seawater are generally stable due to continuous mixing, but can decrease in polluted or stagnant zones. Turbidity in marine environments is usually low in open oceans but can increase near coastal regions due to sediment runoff, industrial discharge, or human activities. High turbidity can affect photosynthesis and marine life visibility.

Marine pollution is a major concern affecting water quality. Oil spills, plastic waste, chemical discharge, and agricultural runoff introduce harmful substances into the ocean, disrupting ecosystems and threatening biodiversity. Monitoring marine water quality is challenging due to its vastness, but it is essential for sustainable fisheries and environmental conservation.

1.9 Water Quality in Aquaculture Systems

Water quality management in aquaculture systems is essential for ensuring optimal fish growth, health, and productivity. Unlike natural ecosystems, aquaculture environments are artificially controlled, making it the responsibility of the farmer or system to maintain ideal conditions. Poor water quality can lead to disease outbreaks, reduced growth rates, and economic losses.

Temperature is one of the most important parameters, as it directly affects fish metabolism and feeding behavior. Each species has an optimal temperature range, and deviations can cause stress. Dissolved oxygen must be maintained at sufficient levels, often through aeration systems, to support fish respiration in densely stocked environments.

Turbidity and water clarity are important for maintaining a healthy environment. Excess feed, fish waste, and microbial activity can increase turbidity and degrade water quality. Additionally, parameters such as ammonia, nitrite, and nitrate levels must be monitored, as these toxic compounds can accumulate due to waste decomposition.

pH levels also play a crucial role, as extreme acidity or alkalinity can harm fish and affect biological processes. In aquaculture systems, regular monitoring and control measures such as filtration, aeration, and water exchange are used to maintain balance.Modern aquaculture increasingly relies on sensor-based monitoring and automated systems to track water quality parameters in real time. These smart systems improve efficiency, reduce manual effort, and enable early detection of unfavorable conditions, ensuring sustainable and productive fish farming practices.

2. LITERATURE REVIEW

The development of smart aquaculture systems has gained significant attention in recent years due to the increasing demand for efficient and sustainable fish farming practices. Traditional aquaculture methods rely heavily on manual monitoring, which is time-consuming, labor-intensive, and prone to human error. To address these limitations, researchers have explored the integration of Internet of Things (IoT), sensor networks, and artificial intelligence (AI) for automated monitoring and management of aquatic environments.

Several studies have focused on IoT-based water quality monitoring systems that utilize sensors to measure parameters such as temperature, pH, turbidity, and dissolved oxygen. These systems typically use microcontrollers like Arduino or ESP-based modules to collect and transmit data to cloud platforms or local servers. Such approaches enable real-time monitoring and remote access, improving decision-making and reducing the risk of fish mortality due to unfavorable environmental conditions.

In addition to sensor-based monitoring, computer vision techniques have been increasingly applied in aquaculture. Image processing methods using tools like OpenCV have been used to observe fish behavior, detect abnormalities, and estimate fish count. However, traditional image processing methods often lack accuracy and robustness in complex underwater environments.

Recent advancements in deep learning have led to the adoption of object detection models such as YOLO (You Only Look Once) for real-time fish detection and tracking. YOLO-based systems provide high accuracy and speed, making them suitable for continuous monitoring applications. Researchers have demonstrated the effectiveness of YOLO models in detecting marine species, analyzing fish movement, and improving feeding strategies.

Some modern systems combine IoT and AI technologies to create intelligent monitoring platforms. These systems integrate sensor data with visual data to provide a comprehensive understanding of aquaculture environments. Web-based dashboards and mobile applications are often used to visualize real-time data, enabling users to monitor conditions remotely.

Despite these advancements, challenges such as system cost, real-time processing limitations, and integration complexity still exist. Many existing systems either focus only on water quality monitoring or only on fish detection, lacking a unified solution.

According to Sameerchand Pudaruth et al. (2021), the identification of fish species remains a significant challenge for the general public, as it often requires access to expert knowledge. To address this issue, the authors developed a smartphone-based application, *SuperFish*, aimed at identifying fish species commonly found in the lagoons, coastal regions, estuaries, and outer reef zones of Mauritius. The study utilized a dataset comprising 1,520 images representing 38 different fish species, with 80% of the data allocated for training, 10% for validation, and the remaining 10% for testing.

The methodology involved several image preprocessing steps, including grayscale conversion and the application of Gaussian blur to reduce noise. A thresholding technique was then used to separate the fish from the background, enabling contour detection and feature extraction.

Multiple machine learning algorithms such as k-Nearest Neighbors (kNN), Support Vector Machines (SVM), decision trees, random forests, and neural networks were evaluated to determine the most effective classifier. The results indicated that the kNN algorithm achieved an accuracy of 96%, while a deep learning model implemented using the TensorFlow framework achieved a higher accuracy of 98%.

The findings demonstrate that both traditional machine learning and deep learning approaches can be effectively used for fish species recognition. However, the study highlights certain limitations, including challenges related to partial occlusions and variations in fish pose. The authors suggest that future work should focus on expanding the dataset and employing more advanced deep learning architectures to improve robustness and accuracy.

According to Catarina N. S. Silva et al. (2022), the increasing use of citizen science platforms, social media, and smartphone applications has led to the generation of large volumes of georeferenced fish images. This development presents significant opportunities for biodiversity and ecological research, particularly in the domain of recreational and small-scale fisheries, which are often considered data-deficient despite their ecological and societal importance. The authors emphasize that artificial intelligence (AI) and computer vision technologies can play a crucial role in automating the analysis of such data by enabling fish species identification and size estimation.

The study proposes a scalable and open-source framework designed to facilitate the storage, handling, annotation, and automatic classification of large-scale image datasets. The framework is built using the TensorFlow Lite Model Maker library and incorporates techniques such as data augmentation and transfer learning applied to various convolutional neural network (CNN) models. The methodology focuses on providing a cost-effective and user-friendly solution that reduces the need for extensive programming expertise, thereby making AI tools more accessible to researchers and practitioners in the fisheries sector.

To demonstrate the effectiveness of the proposed system, the authors applied the framework to a sample dataset of fish images collected through a recreational fishing smartphone application. The results indicate that the framework is capable of supporting region-specific fish species identification models, which can be further extended into hierarchical classification systems for broader applications.

However, the study highlights several limitations, including the current focus of many AI-based fisheries applications on commercial sectors and specific species, as well as the lack of publicly available and generalized tools. Additionally, the effectiveness of the system depends on the quality and diversity of the dataset. The authors suggest that future research should focus on expanding datasets and improving model generalization to support large-scale ecological monitoring and fisheries management.

According to John Michael De Jesus et al. (2019), most digital image processing techniques for fish counting are effective only under clear water conditions, limiting their applicability in real-world aquaculture environments. The study investigates the relationship between water turbidity and the accuracy of fish counting using computer vision techniques. The proposed system utilizes OpenCV-based video analysis algorithms to estimate fish population in fish pens, specifically targeting Nile Tilapia (*Oreochromis niloticus*), for applications in small-scale fisheries and monitoring by the Bureau of Fisheries and Aquatic Resources (BFAR) in the Philippines.

The methodology involved deploying the system in an actual aquaculture setting located in Los Baños, Laguna, where calibration and testing were conducted over a period from December 2018 to March 2019. The system performed fish detection and counting from video data under varying turbidity conditions, enabling a comparative analysis with manual counting methods. The results indicated that there was no significant difference between the automated system and manual counting, demonstrating the reliability and effectiveness of the proposed approach.

However, the study acknowledges limitations related to varying water clarity, which can still impact detection accuracy. Despite this, the system proves to be a viable alternative to manual counting, offering a more efficient and scalable solution for fish monitoring and data acquisition in aquaculture environments.

According to Siti Nurulain Mohd Rum et al. (2021), underwater image processing plays a crucial role in fish species identification, which is important for biologists, researchers, fishermen, and educational purposes. The study highlights that Malaysia alone has over 200 species of freshwater fish, creating a need for efficient and automated identification systems. While previous research has focused mainly on saltwater species and specific user groups, this work aims to address the gap by developing a system for freshwater fish identification.

The authors proposed a prototype system named *FishDeTec*, which utilizes image processing techniques for detecting and classifying freshwater fish species found in Malaysia. The methodology is based on a deep learning approach using the VGG16 convolutional neural network (CNN), a well-established model for large-scale image classification tasks. The system processes fish images and extracts relevant features to perform accurate classification.

The experimental results demonstrate that the proposed model achieves promising performance in identifying fish species, indicating the effectiveness of deep learning-based approaches in underwater image analysis. However, the study is limited by the scope of the dataset and the focus on specific regional species. The authors suggest that future improvements could include expanding the dataset and enhancing the model to handle more diverse environmental conditions and species variations.

According to Pradnya Narvekar et al. (2022), underwater camera systems are widely used in marine ecology for monitoring aquatic species; however, the large volume of data generated requires efficient automated processing techniques. The study addresses the challenges of fish species classification in natural environments, where factors such as noise, illumination variation, and complex backgrounds significantly affect detection accuracy.

The authors proposed a two-step deep learning approach for fish detection and classification without the need for pre-filtering. The methodology utilizes transfer learning to overcome the limitation of small training datasets. Initially, the detection model is trained using the ImageNet dataset, followed by training the classification model using the Fish4Knowledge dataset. Both models are then fine-tuned using images of temperate fish species. The pre-trained weights are leveraged to improve model performance during post-training.

The results demonstrate that the proposed approach achieves a high accuracy of 99.27% during pre-training. In post-training, the precision values obtained were 83.68% and 87.74% with and without image augmentation, respectively, indicating that the model performs effectively under real-world conditions. However, the study is limited by dataset size and environmental variability, which can impact performance. The authors suggest that increasing the dataset size and diversity would further enhance the robustness and accuracy of the system.

According to Sonali Guhe et al. (2023), computer vision techniques have been increasingly applied for fish detection using underwater imagery. The study proposes a fish detection system developed using Python and the OpenCV library, focusing on detecting fish in aquatic environments through image processing methods.

The methodology involves several key steps, including image filtering to reduce noise, segmentation to isolate fish from the background, and feature extraction to identify relevant characteristics for detection. These techniques enable the system to process underwater images and detect the presence of fish effectively.

The results demonstrate that computer vision-based approaches have significant potential for fish detection in real-world aquatic environments. However, the study is limited by the reliance on traditional image processing techniques, which may not perform well under complex conditions such as poor visibility, varying lighting, and occlusions. The authors suggest that integrating advanced methods such as deep learning could further improve detection accuracy and system robustness.

According to Manikandan L R et al. (2021), deep learning techniques have been effectively applied for fish detection and recognition in underwater environments. The study proposes a system based on Region-based Convolutional Neural Networks (R-CNN) combined with OpenCV tools to detect and classify fish species from images captured in natural lake environments.

The methodology utilizes the selective search algorithm to generate region proposals, from which the top candidate regions are passed to a convolutional neural network for classification. Fish images are first acquired using a custom-built image acquisition system, followed by preprocessing and labeling before being used for training the R-CNN model. In addition to detection, the system also performs image segmentation to extract morphological features of fish, which can be useful for ecological analysis.

The results indicate that the proposed system achieves an accuracy of approximately 85% in fish detection and recognition tasks. This demonstrates the effectiveness of deep learning approaches in handling complex underwater environments. However, the study is limited by moderate accuracy and potential challenges such as computational complexity and sensitivity to environmental variations. The authors suggest that further improvements can be achieved by using larger datasets and more advanced deep learning architectures to enhance detection performance and reliability.

According to Ian C. Navotas et al. (2018), the increasing demand for fish due to its high nutritional value necessitates accurate methods for assessing fish freshness, as manual evaluation by non-experts can lead to incorrect judgments and potential health risks. The study proposes an

Android-based application that automatically identifies commonly consumed fish species in the Philippines, such as milkfish, round scad, and tilapia, while also determining their freshness levels.

The methodology employs image processing techniques combined with an artificial neural network to analyze visual features of fish, specifically focusing on the RGB values of the eyes and gills. The system classifies freshness into five levels, ranging from stale to fresh, and estimates the remaining shelf life. The model is trained using a feedforward neural network with a dataset comprising 30 fish samples per species, generating approximately 800 images each for the eyes and gills through iterative learning.

The results demonstrate that the proposed system achieves acceptable performance in both fish identification and freshness classification tasks. However, the study is limited by the relatively small dataset and the focus on only a few species. The authors suggest that expanding the dataset and incorporating more advanced learning techniques could improve the accuracy and generalization of the system.

According to Faizan Farooq Khan et al. (2023), aquatic biodiversity is increasingly threatened by factors such as overfishing, climate change, and coastal development, necessitating the development of automated monitoring systems for effective conservation. However, the advancement of such systems is hindered by the lack of large-scale, diverse datasets that can support accurate species recognition and ecological analysis.

To address this limitation, the authors introduced *FishNet*, a large-scale dataset comprising 94,532 images representing 17,357 aquatic species, organized according to biological taxonomy including order, family, genus, and species. The dataset is designed to support multiple tasks such as fish classification, detection, and functional trait prediction. In addition, the study establishes three benchmark tasks aligned with ecological research needs to facilitate the development and evaluation of machine learning models.

The methodology emphasizes creating a comprehensive and structured dataset that can be used with deep learning techniques to improve species recognition and ecological understanding. The results suggest that such a large-scale dataset significantly enhances the potential for developing generalized and accurate aquatic monitoring systems.

However, the study is primarily focused on dataset development rather than deployment in real-time systems. The effectiveness of models trained on this dataset depends on proper implementation and computational resources. The authors suggest that the availability of FishNet will promote further research and enable the creation of more robust and scalable solutions for aquatic ecosystem monitoring and biodiversity conservation.

According to Sameerchand Pudaruth et al. (2021), identifying fish species is a challenging task for the general public due to the need for specialized knowledge. To address this issue, the study presents a smartphone-based application designed to identify fish species commonly found in the coastal and lagoon regions of Mauritius. The dataset used in the study consists of 1,520 images representing 38 fish species, with standard data splitting into training, validation, and testing sets.

The methodology involves multiple image preprocessing steps, including grayscale conversion and Gaussian blur for noise reduction, followed by thresholding to separate fish from the background. Contour detection is then applied to extract features, which are used for classification. Various machine learning algorithms such as k-Nearest Neighbors (kNN), Support Vector Machines (SVM), decision trees, random forests, and neural networks were evaluated. Among these, the kNN algorithm achieved an accuracy of 96%. Additionally, a deep learning model implemented using the TensorFlow framework achieved a higher accuracy of 98%.

The results demonstrate the effectiveness of combining traditional image processing techniques with machine learning and deep learning approaches for fish species identification. However, the study is limited by dataset size and challenges such as partial occlusion and pose variation. The authors suggest that future work should focus on expanding the dataset and employing more advanced deep learning models to improve robustness and accuracy.

According to Thi Thu Em Vo et al. (2021), smart aquaculture has emerged as a key trend in modern aquaculture, focusing on automation, intelligence, and sustainability. The study highlights how advanced technologies, particularly machine learning and computer vision, have significantly contributed to improving aquaculture practices by reducing labor requirements, increasing production efficiency, and minimizing environmental impact.

The paper presents a comprehensive review of approximately 100 research articles published over the past decade, focusing on the application of machine learning in aquaculture. The methodologies discussed include various AI-based approaches for tasks such as fish size and weight estimation, grading, disease detection, and species classification. The study emphasizes that machine learning models are capable of learning patterns from large datasets, enabling more accurate and efficient monitoring systems.

The findings suggest that integrating intelligent technologies into aquaculture systems enhances productivity and supports sustainable development. However, the study also identifies challenges such as the need for large datasets, high computational requirements, and the complexity of implementing these technologies in real-world environments. The authors recommend further research in developing cost-effective, scalable, and user-friendly smart aquaculture systems that can be widely adopted.

According to Priyanshi Singh et al. (2025), smart aquaculture integrates advanced technologies such as artificial intelligence (AI), machine learning, Internet of Things (IoT), and data analytics to overcome the limitations of traditional fish farming and promote sustainable practices. The study highlights key challenges in conventional aquaculture, including water quality management, disease detection, feed optimization, and high labor dependency.

The methodology discussed involves the use of sensor networks, automation systems, computer vision techniques, and blockchain technology to enable real-time monitoring, predictive analytics, and transparent supply chain management. Technologies such as Recirculatory Aquaculture Systems (RAS), smart sensors, and automated feeding mechanisms are emphasized as critical components in improving efficiency and fish health.

The findings indicate that smart aquaculture systems significantly enhance productivity, optimize resource utilization, and reduce environmental impact. Furthermore, the study envisions future intelligent fish farms that can operate with minimal human intervention. However, limitations such as high implementation costs, technical complexity, and the requirement for skilled personnel are identified. The authors suggest that overcoming these challenges will be essential for the widespread adoption of smart aquaculture and for meeting the increasing global demand for seafood in a sustainable manner.

According to Abdul Azeem et al. (2024), the Internet of Things (IoT) has significantly transformed various industries, including aquaculture, by enabling intelligent monitoring and automation. The study focuses on addressing key challenges in aquaculture such as water quality management and disease prevention, which require continuous monitoring and timely intervention.

The proposed system integrates multiple sensors, including water level, temperature, and pH sensors, connected to an Arduino Uno R3 microcontroller along with an ESP8266 Wi-Fi module. The methodology involves real-time data acquisition and transmission to the ThingSpeak IoT platform, allowing remote monitoring through web and mobile interfaces. Additionally, the system incorporates automation by controlling a DC water pump based on sensor readings, ensuring optimal water conditions without manual intervention.

The results indicate that the system provides an efficient and cost-effective solution for real-time monitoring and management of aquaculture environments. However, the study is limited by its reliance on stable internet connectivity and basic sensor parameters, which may not fully capture complex environmental conditions. The authors suggest that future improvements could include the integration of additional sensors and advanced analytics to enhance system performance and scalability.

According to Zhuhua Hu et al. (2020), aquaculture plays a vital role in the agricultural economy, but traditional practices often rely on subjective experience, leading to inefficiencies and potential risks. The study emphasizes that maintaining optimal water quality is critical for ensuring productivity and sustainability in aquaculture systems. With advancements in technology, smart aquaculture has emerged as a solution to enable intelligent monitoring and management of aquatic environments.

The paper provides a comprehensive review of research conducted over the past two decades, focusing on four key areas: water quality data acquisition and preprocessing, prediction of water quality parameters, analysis of fish morphological characteristics, and recognition of fish behavioral patterns. The methodology discussed includes the use of sensor-based systems, data-driven models, and machine learning algorithms to monitor and predict environmental changes, as well as to analyze fish behavior for indirect assessment of water quality.

The findings highlight that integrating environmental monitoring with fish behavior analysis can significantly improve aquaculture management by enabling real-time prediction, warning, and risk control. However, the study also identifies limitations such as the complexity of modeling interactions between water quality and fish behavior, and the need for more accurate and scalable algorithms. The authors suggest that future research should focus on developing integrated, intelligent systems that combine multiple data sources to enhance the efficiency, sustainability, and ecological balance of aquaculture systems.

According to Ritish Biswas et al., water quality monitoring is essential for ensuring healthy fish growth and preventing critical conditions in aquaculture systems. The study emphasizes that key parameters such as temperature, dissolved oxygen, and pH significantly influence fish health, feeding efficiency, growth rate, and overall productivity. Poor or fluctuating environmental conditions can lead to stress and disease outbreaks in fish populations.

The proposed system is an IoT-based smart aquaculture monitoring solution that integrates sensors to continuously monitor water quality parameters. The methodology involves real-time data collection and control mechanisms, including an automated feeding system that replaces traditional manual feeding practices. The system is designed to be controlled through an Android application, enabling remote operation and improved management efficiency.

The results indicate that the system enhances monitoring accuracy and reduces manual effort, contributing to better fish health and productivity. However, the study is limited by its focus on a few environmental parameters and basic automation features. The authors suggest that future improvements could include the integration of additional sensors and advanced analytics to provide a more comprehensive and intelligent aquaculture management system.

According to Prathiba P et al. (2019), aquaculture remains a technologically underdeveloped sector compared to other domains such as agriculture, leading to challenges like delayed response in water quality management, resource wastage, and production losses. The study proposes a smart aquaculture system based on wireless sensor networks and Internet of Things (IoT) technologies to address these issues.

The methodology involves deploying distributed sensors to monitor key water quality parameters in real time. The system integrates multiple functionalities, including automatic fish feeding, water cleaning processes triggered by pH fluctuations, and fish mortality detection. Data collected from the sensors is transmitted through IoT platforms, enabling remote monitoring and information sharing across different stakeholders and organizations.

The results indicate that the system improves the efficiency of aquaculture management by enabling timely interventions, optimizing resource utilization, and enhancing the overall living conditions of aquatic organisms. However, the study is limited by implementation complexity and dependency on network infrastructure. The authors suggest that further development should focus on improving system scalability, reliability, and integration with advanced data analytics for better decision-making.

According to Rishika M S, aquaculture plays a crucial role in ensuring global food security by providing protein-rich nutrition, supporting livelihoods, and maintaining ecosystem sustainability. However, increasing anthropogenic activities and greenhouse gas emissions pose significant threats to aquaculture systems, necessitating the adoption of advanced technological solutions.

The study introduces the concept of Climate-Smart Aquaculture, which focuses on integrating innovative technologies and sustainable practices to enhance resilience against climate change. The methodology highlights the use of advanced systems such as Recirculatory Aquaculture Systems (RAS), Integrated Multi-Trophic Aquaculture (IMTA), and artificial intelligence-based monitoring for efficient farm management. These approaches enable real-time monitoring, improved resource utilization, and adaptation to changing environmental conditions.

The findings suggest that climate-smart aquaculture promotes three primary objectives: increasing production and revenue, adapting to climate change, and mitigating its impacts. However, challenges such as high implementation costs, technological complexity, and the need for skilled expertise limit widespread adoption. The study emphasizes that further research and development are required to make these technologies more accessible and scalable for sustainable aquaculture development.

According to Ashutosh Danve et al., smart automation has become a significant trend in aquaculture due to advancements in science and technology, enabling intelligent and data-driven farming practices. The study highlights that smart aquaculture systems can perform real-time monitoring, prediction, warning, and risk control of environmental parameters, while also analyzing fish behavior to infer ecological changes within aquaculture systems.

The methodology focuses on the integration of advanced technologies such as artificial intelligence, Internet of Things (IoT), robotics, edge computing, 5G communication, and vision-based systems. These technologies support automated feeding, water quality management, fish health assessment, and harvesting processes. In particular, robotic systems equipped with vision sensors can identify, sort, and harvest fish efficiently, reducing manual labor and improving operational accuracy. Additionally, automated data collection and analysis systems enable better decision-making and optimization of aquaculture processes.

The findings indicate that smart automation significantly enhances productivity, reduces labor dependency, and promotes sustainable aquaculture practices. However, the study also identifies limitations such as high initial investment costs, technical complexity, potential job displacement, and ethical concerns related to animal welfare and environmental impact. The authors suggest that further development and adoption of these technologies are necessary to achieve efficient, sustainable, and economically viable aquaculture systems.

According to S. Archana et al. (2025), digital transformation is significantly reshaping fisheries and aquaculture through the integration of Internet of Things (IoT), advanced sensor systems, and artificial intelligence (AI)-driven analytics. The study highlights that smart fishing technologies enable real-time environmental monitoring, precision aquaculture, automated catch analysis, and data-driven decision-making in fisheries management.

The methodology discussed involves the use of IoT networks incorporating physicochemical sensors, optical and acoustic systems, electronic monitoring cameras, and satellite data streams. These systems collect large-scale data, which is then analyzed using machine learning algorithms to optimize fishing operations, detect bycatch, monitor vessel compliance, and predict aquaculture system health.

The findings indicate that the integration of IoT and AI enhances efficiency, sustainability, and governance in modern fisheries. However, the study identifies several challenges, including data interoperability issues, infrastructure limitations, ethical concerns, and policy gaps. The authors suggest that future research should focus on overcoming these barriers to achieve effective digital transformation and sustainable development in global fisheries through “blue digitalization.”

According to Yanbo Wang et al., smart fishing rod systems incorporating vibration sensors represent an innovative advancement in angling technology. The study focuses on the use of piezoelectric-based vibration sensors integrated with flex sensors and Bluetooth communication to detect fish bites in real time and enhance the fishing experience.

The methodology involves embedding sensors within the fishing rod to capture subtle vibrations caused by fish activity. These sensors are capable of distinguishing between actual fish bites and external disturbances, thereby reducing false alarms. The system allows calibration of sensitivity and detection range to adapt to different fishing conditions. Data from the sensors is transmitted via Bluetooth to connected devices, enabling real-time monitoring and feedback for the user.

The findings indicate that such smart systems significantly improve bite detection accuracy and increase catch efficiency for both novice and experienced anglers. However, the study is limited by its focus on individual fishing applications rather than large-scale aquaculture systems. The authors suggest that future advancements could further enhance sensor precision and integration with more advanced monitoring technologies for broader applications.

3. Methodology

The methodology of this project describes the systematic approach used to design and implement a smart fish detection and water monitoring system by integrating embedded hardware, IoT communication, and artificial intelligence techniques. The system is developed using an ESP32-S3 CAM module for data acquisition and a Python-based server using the Flask framework for processing and visualization.

The ESP32-S3 collects real-time environmental data from multiple sensors, including water temperature, surface temperature, water level, pressure, and turbidity. It also captures live video streams of the aquatic environment using an onboard camera. This data is transmitted over a Wi-Fi network to a local server for further processing.

On the server side, a Flask-based web application is used to handle data communication and visualization. The live video stream received from the ESP32 is processed using OpenCV, and object detection is performed using a pre-trained YOLO (You Only Look Once) deep learning model. This enables real-time identification and tracking of fish in the captured frames.

Sensor data is periodically fetched from the ESP32 and integrated with the detection results to provide a unified monitoring system. The processed information is displayed on a dynamic web dashboard, allowing users to observe both environmental parameters and fish activity in real time.

This methodology ensures efficient data acquisition, intelligent processing, and user-friendly visualization, resulting in a robust and automated system for aquaculture monitoring. By combining IoT and AI technologies, the system enhances accuracy, reduces manual effort, and enables better decision-making in aquatic environments.

3.1 Block Diagram Description

The block diagram represents the overall architecture of the smart fish detection and water monitoring system. At the core of the system is the ESP32-S3 microcontroller, which acts as the main processing and control unit. It interfaces with multiple sensors and modules to collect environmental data and manage system operations.

The JNS-SR04T ultrasonic sensor is used to measure the water level or distance by calculating the time taken for ultrasonic waves to reflect from the surface. The DS18B20 temperature sensor is used to measure water temperature accurately, while the DHT11 sensor measures ambient temperature and humidity. The BMP180 sensor is used to measure atmospheric pressure and altitude, which helps in environmental analysis.

A turbidity sensor (water quality sensor) is used to determine the clarity of water, which is an important parameter for fish health. All these sensors continuously send data to the ESP32-S3 for processing.

The ESP32-S3 CAM module also includes a camera that captures real-time images or video of the aquatic environment. A power module is used to supply stable voltage to all components, ensuring reliable operation.

The processed data is then transmitted via Wi-Fi to external devices such as a laptop or mobile device, where it can be visualized through a web interface. This architecture enables real-time monitoring and intelligent decision-making.

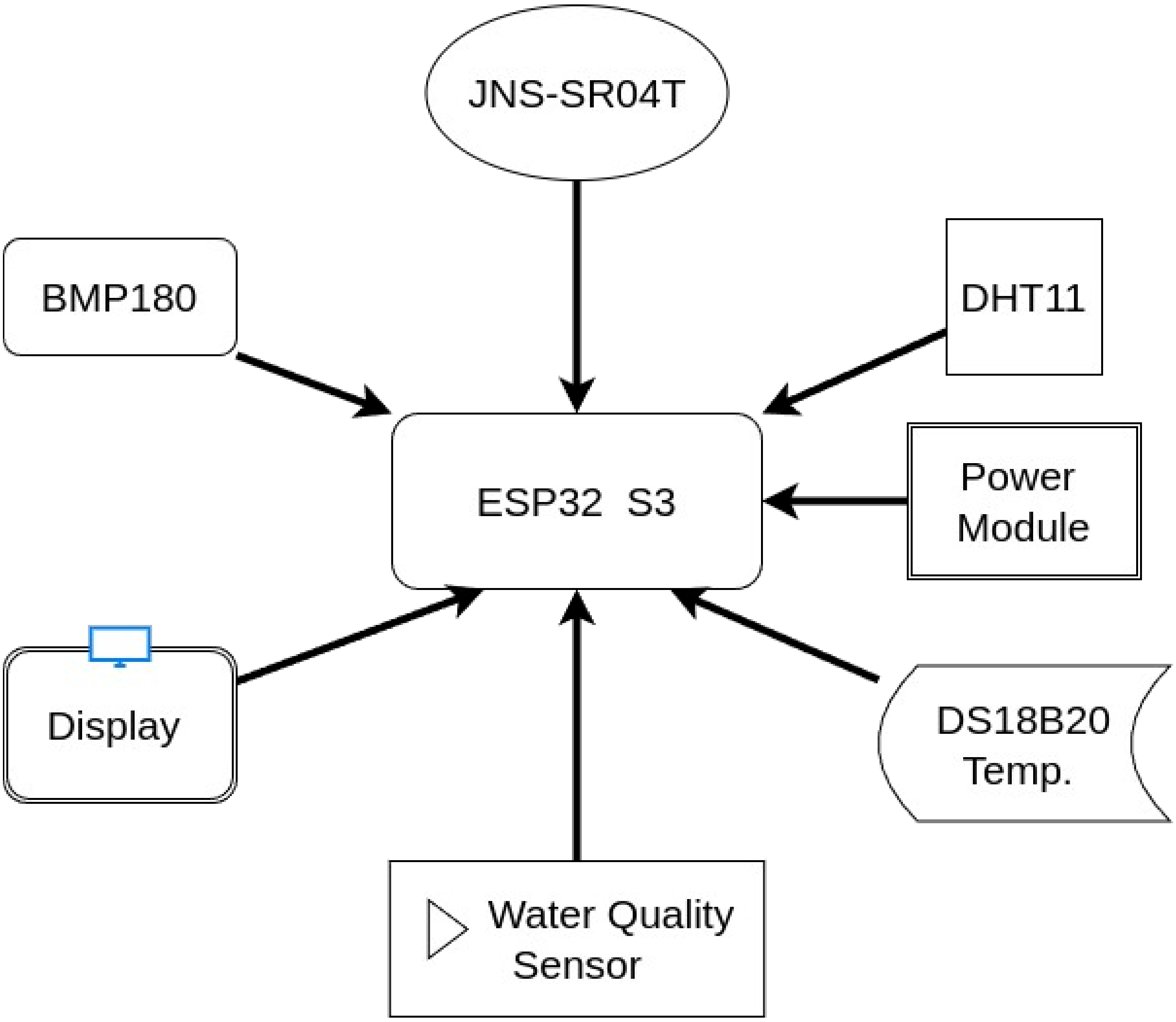

Fig-3.1: Block Diagram

3.2 Workflow Description

The workflow of the system illustrates the step-by-step operation from data collection to user visualization. Initially, the sensors connected to the ESP32-S3 collect environmental parameters such as water temperature, surface temperature, water level, turbidity, and pressure. At the same time, the onboard camera captures live video of the aquatic environment.

Once the data is collected, the ESP32-S3 processes the sensor readings and streams the video through a Wi-Fi network. The data is transmitted using HTTP protocols to a local server system.

On the receiving side, a Python-based Flask server processes the incoming data. The video stream is decoded using OpenCV, and a YOLO deep learning model is applied to detect and identify fish in real time. The detection results, along with sensor data, are combined and updated continuously.

The processed data is then displayed on a web-based dashboard, which includes live video feed with detection overlays, sensor readings, and system status indicators. Users can access this dashboard through a laptop or mobile device.

The entire workflow operates continuously in a loop, enabling real-time monitoring, early detection of issues, and improved management of aquaculture environments.

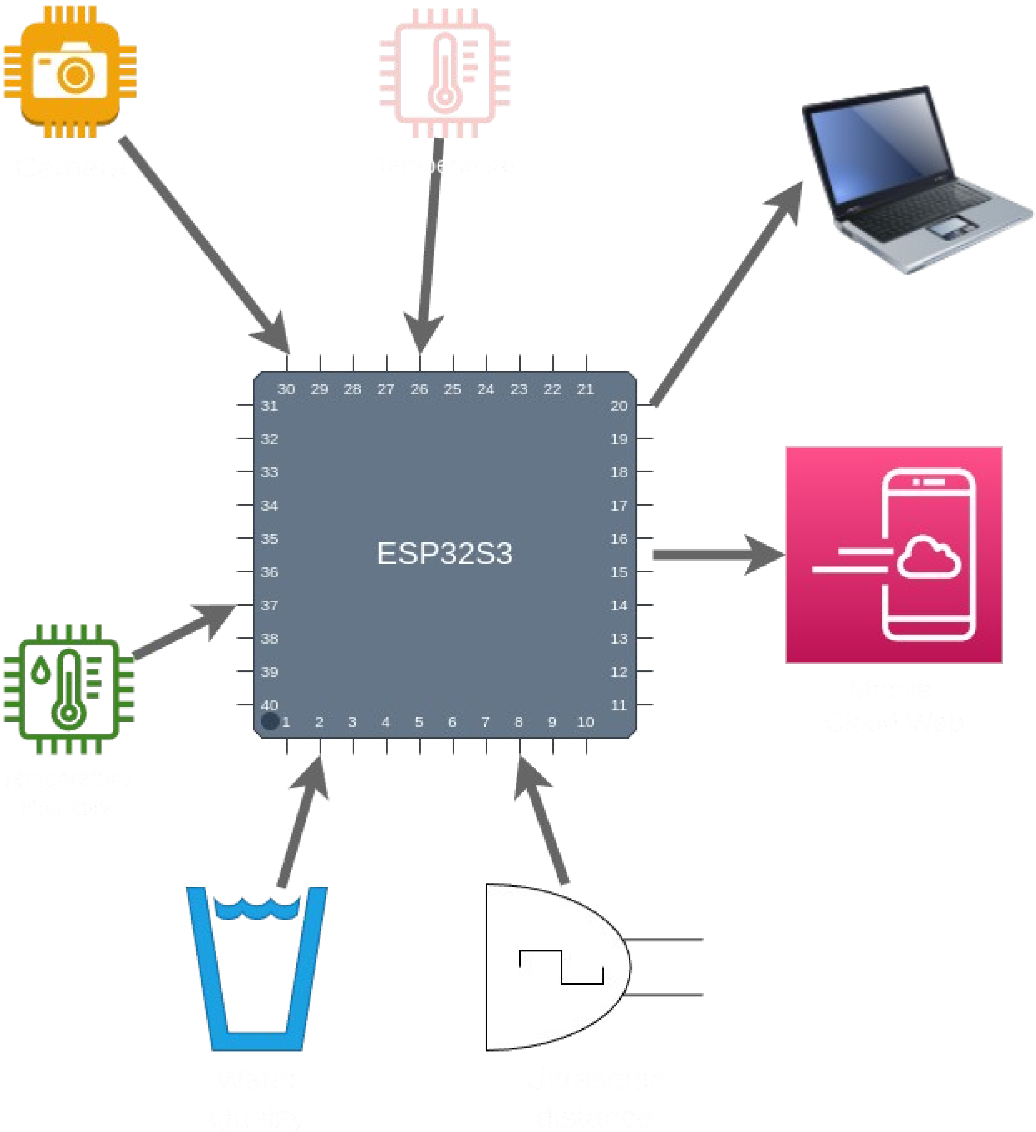

Fig-3.2: Workflow Diagram

3.3 Software Description

The software component of the proposed smart fish detection and water monitoring system plays a crucial role in data processing, communication, visualization, and intelligent analysis. The system integrates multiple software tools and frameworks, including embedded programming, web development, and artificial intelligence libraries, to achieve real-time monitoring and automation.

3.3.1 Embedded Programming (Arduino IDE)

The ESP32-S3 microcontroller is programmed using the Arduino IDE, which provides a simple and efficient environment for developing embedded applications. The firmware is responsible for interfacing with various sensors such as DS18B20, DHT11, ultrasonic sensor, BMP180, and turbidity sensor. It collects real-time environmental data, processes basic calculations, and manages communication over Wi-Fi. The ESP32 also handles camera initialization and streams image data through HTTP protocols.

3.3.2 Python and Flask Framework

A Python-based server is implemented using the Flask framework to handle backend processing and communication. Flask acts as a lightweight web server that receives data from the ESP32-S3 and provides API endpoints for accessing sensor data and video streams. It manages routing, data handling, and integration between different components of the system. The server continuously fetches sensor data and processes incoming video frames for further analysis.

3.3.3 OpenCV (Computer Vision Library)

OpenCV is used for real-time image processing and video stream handling. It decodes the incoming MJPEG stream from the ESP32 camera and converts it into frames suitable for analysis. OpenCV is also used to draw bounding boxes, labels, and overlays on detected objects, enhancing visualization and interpretation of results.

3.3.4 YOLO (Deep Learning Model)

The system uses a YOLO (You Only Look Once) model for object detection, specifically trained to detect fish or marine objects. YOLO is a real-time object detection algorithm known for its speed and accuracy. It processes each video frame and identifies objects by generating bounding boxes and class labels. The detection results are used to monitor fish presence and activity in the aquatic environment.

3.3.5 Frontend Web Technologies

The user interface is developed using HTML, CSS, and JavaScript. The web dashboard provides a real-time visualization of sensor data and live video feed. Chart.js is used for plotting graphs such as water level trends, while JavaScript handles dynamic updates and asynchronous data fetching from the Flask server. The interface is designed to be responsive and accessible from both desktop and mobile devices.

3.3.6 Communication Protocols

The system uses HTTP-based communication for data exchange between the ESP32-S3 and the Flask server. Sensor data is transmitted in JSON format through API endpoints, while the video stream is delivered using MJPEG streaming. This approach ensures efficient and real-time data transfer over a Wi-Fi network.

3.4 ESP32-S3 CAM Module

The ESP32-S3 CAM module is the core hardware component of the proposed smart fish detection and water monitoring system. It is a powerful microcontroller unit developed by Espressif Systems, featuring integrated Wi-Fi and enhanced processing capabilities suitable for embedded vision and IoT applications. The module is specifically designed to support camera-based applications while maintaining low power consumption and compact size.

The ESP32-S3 is built on a dual-core processor architecture, which enables efficient multitasking and real-time data processing. It includes sufficient GPIO pins to interface with various sensors such as temperature sensors, ultrasonic sensors, turbidity sensors, and pressure sensors. In addition, it supports communication protocols such as I2C, SPI, and UART, allowing seamless integration with multiple hardware components.

One of the key features of the ESP32-S3 CAM module is its compatibility with camera sensors such as the OV3660. This allows the system to capture images and stream video in real time. The module supports JPEG image compression, which reduces data size and enables efficient transmission over Wi-Fi networks. This feature is particularly important for real-time monitoring applications where bandwidth optimization is required.

The built-in Wi-Fi capability of the ESP32-S3 enables wireless communication with external devices such as laptops, mobile phones, or cloud servers. In this project, the ESP32-S3 acts as a web server that transmits sensor data and live video streams to a Flask-based application for further processing and visualization.

The module also supports external PSRAM, which enhances its ability to handle large image buffers and improves overall performance during video streaming and image processing tasks. This makes it suitable for applications involving computer vision and real-time monitoring.

Furthermore, the ESP32-S3 CAM module operates at low power and can be powered using a regulated power supply, making it suitable for continuous operation in aquaculture environments. Its compact design, processing capability, and wireless connectivity make it an ideal choice for integrating sensing, processing, and communication in a single platform.

3.5 Ultrasonic Sensor (JNS-SR04T)

The JNS-SR04T ultrasonic sensor is used in this project to measure the distance between the sensor and the water surface, which helps in determining the water level in the pond or tank. It is a waterproof ultrasonic distance sensor, making it highly suitable for outdoor and aquatic environments where exposure to moisture is common.

The sensor operates on the principle of ultrasonic wave propagation. It consists of a transmitter and a receiver. The transmitter emits high-frequency ultrasonic sound waves (typically around 40 kHz), which travel through the air and reflect back when they encounter an object, such as the water surface. The receiver detects the reflected signal, and the time taken for the wave to return is used to calculate the distance.

The distance is calculated using the time-of-flight principle, based on the speed of sound in air. The ESP32-S3 microcontroller sends a trigger pulse to the sensor, which initiates the transmission of ultrasonic waves. When the echo signal is received, the duration of the pulse is measured, and the distance is computed using a mathematical formula.

The JNS-SR04T sensor is capable of measuring distances typically ranging from 20 cm to 600 cm with reasonable accuracy. Its waterproof probe design ensures reliable performance in humid or wet conditions, unlike standard ultrasonic sensors that are sensitive to moisture.

In this system, the sensor is mounted above the water surface, and the measured distance is used to estimate the water level. By comparing the measured distance with the known height of the tank or pond, the system can determine whether the water level is within the desired range.

The sensor provides a digital output in the form of a pulse signal, which is easily interfaced with the ESP32-S3 through GPIO pins. Its low cost, ease of use, and robustness make it an ideal choice for real-time water level monitoring in aquaculture applications.

3.6 DS18B20 Temperature Sensor

The DS18B20 is a digital temperature sensor used in this project to measure the water temperature accurately. It is widely used in embedded and IoT applications due to its high precision, reliability, and ease of interfacing. The sensor is particularly suitable for aquatic environments because it is available in a waterproof probe form, making it ideal for direct immersion in water.

The DS18B20 operates using the One-Wire communication protocol, which allows data transmission through a single data line along with ground. This reduces wiring complexity and enables multiple sensors to be connected to the same microcontroller pin if required. Each DS18B20 sensor has a unique 64-bit serial code, allowing multiple devices to operate on the same bus without conflict.

The sensor provides digital temperature readings with a high degree of accuracy, typically ±0.5°C over a wide temperature range of -55°C to +125°C. In this project, it is primarily used within the range suitable for aquaculture, where maintaining optimal water temperature is essential for fish health and growth.

The DS18B20 measures temperature internally and converts the reading into a digital signal, eliminating the need for external analog-to-digital conversion. The measured data is then transmitted to the ESP32-S3 microcontroller, where it is processed and displayed on the monitoring system.

In the proposed system, the DS18B20 sensor is used to continuously monitor the water temperature, which is a critical parameter in aquaculture. Sudden changes in water temperature can stress fish and affect their metabolism, feeding behavior, and overall health. By providing real-time temperature data, the system enables timely intervention and better environmental control.

The sensor is easy to interface with the ESP32-S3 using a pull-up resistor and compatible libraries, making it a cost-effective and efficient solution for temperature monitoring. Its durability, accuracy, and waterproof design make it highly suitable for long-term deployment in water monitoring applications.

3.7 DHT11 Temperature and Humidity Sensor

The DHT11 is a low-cost digital sensor used to measure ambient temperature and humidity. In this project, it is utilized to monitor the surface environmental conditions around the aquaculture system. Understanding atmospheric temperature and humidity is important, as these factors can influence water temperature, evaporation rates, and overall environmental stability.

The DHT11 sensor consists of a thermistor for measuring temperature and a capacitive humidity sensor for detecting moisture levels in the air. It also includes an internal microcontroller that processes the measured data and outputs it as a digital signal. This eliminates the need for external signal conditioning or analog-to-digital conversion.

The sensor communicates with the ESP32-S3 microcontroller using a single-wire digital communication protocol. It sends data in the form of a timed pulse sequence, which is decoded by the microcontroller to obtain temperature and humidity values. The DHT11 is easy to interface and requires only one data pin along with power and ground connections.

The DHT11 provides temperature measurements in the range of 0°C to 50°C with an accuracy of approximately ±2°C, and humidity measurements from 20% to 90% with an accuracy of ±5%. Although it is not as precise as advanced sensors, it is sufficient for general environmental monitoring in aquaculture systems.

In this system, the DHT11 sensor is used to monitor surface temperature conditions, which can indirectly affect the water environment. Changes in atmospheric temperature can influence water temperature and oxygen levels, thereby impacting fish health. By monitoring these conditions, the system can provide a more comprehensive understanding of the surrounding environment.

The sensor is compact, energy-efficient, and cost-effective, making it suitable for continuous monitoring applications. Its simplicity and ease of integration with the ESP32-S3 make it an ideal choice for basic environmental sensing in this project.

3.8 Turbidity Sensor (Water Quality Sensor)

The turbidity sensor is used in this project to measure the clarity of water, which is an important indicator of water quality in aquaculture systems. Turbidity refers to the presence of suspended particles such as dirt, algae, and organic matter in water. High turbidity levels can reduce light penetration, affect photosynthesis, and negatively impact fish health.

The sensor operates based on the principle of light scattering. It consists of a light source, typically an infrared (IR) LED, and a photodetector. When light passes through water, suspended particles scatter the light. The amount of scattered light detected by the sensor is proportional to the turbidity level of the water. Clear water allows more light to pass through, while turbid water causes more scattering and reduces transmitted light intensity.

In this project, the turbidity sensor provides a digital output indicating the quality of water. The ESP32-S3 reads this signal through a GPIO pin and classifies the water condition as either “Good” or “Bad” based on predefined thresholds. This simplifies real-time monitoring and enables quick decision-making.

Turbidity is a critical parameter in aquaculture, as poor water quality can lead to reduced oxygen levels, increased stress in fish, and higher susceptibility to diseases. Continuous monitoring of turbidity helps in maintaining optimal conditions for fish growth and survival.

The sensor is easy to interface with the ESP32-S3 and requires minimal calibration for basic applications. It is cost-effective and suitable for real-time monitoring systems. By integrating turbidity measurement into the system, the project enhances its capability to assess overall water quality and ensures a healthier aquatic environment.

3.9 BMP180 Pressure Sensor

The BMP180 is a digital barometric pressure sensor used in this project to measure atmospheric pressure and estimate altitude. It is a high-precision, low-power sensor widely used in embedded systems and environmental monitoring applications. In aquaculture systems, pressure and altitude measurements can provide useful insights into environmental conditions and weather changes.

The BMP180 sensor operates based on the piezoresistive principle, where changes in atmospheric pressure cause deformation in an internal sensing element. This deformation results in a change in electrical resistance, which is measured and converted into a digital signal. The sensor includes an internal analog-to-digital converter (ADC) and calibration data, ensuring accurate and stable readings.

The sensor communicates with the ESP32-S3 microcontroller using the I2C communication protocol, which allows efficient data transfer using only two wires: Serial Data (SDA) and Serial Clock (SCL). This makes it easy to integrate with other I2C-based devices in the system.

The BMP180 is capable of measuring pressure in the range of approximately 300 hPa to 1100 hPa, with high accuracy. Using pressure data, the sensor can also estimate altitude based on standard atmospheric models. In this project, altitude calculation helps in understanding environmental conditions and can be useful for calibration and reference purposes.

Although the BMP180 is not directly related to water quality, it contributes to the overall environmental monitoring system by providing additional atmospheric data. Changes in atmospheric pressure can indicate weather variations, which may indirectly affect aquaculture conditions such as temperature and oxygen levels.

The sensor is compact, energy-efficient, and reliable, making it suitable for continuous monitoring applications. Its compatibility with the ESP32-S3 and ease of integration through I2C communication make it an ideal choice for this project.

3.10 YOLO (You Only Look Once) Object Detection Model

YOLO (You Only Look Once) is a state-of-the-art deep learning algorithm used for real-time object detection. In this project, YOLO is utilized to detect and identify fish in the video stream captured by the ESP32-S3 CAM module. Its high speed and accuracy make it suitable for real-time monitoring applications in aquaculture systems.

Unlike traditional object detection methods that perform multiple stages such as region proposal and classification separately, YOLO processes the entire image in a single pass through a convolutional neural network (CNN). This unified approach significantly reduces computation time and enables real-time performance.

The YOLO model divides an input image into a grid and predicts bounding boxes along with class probabilities for each grid cell. Each bounding box contains information about the position and size of the detected object, as well as a confidence score indicating the likelihood of detection. The model then applies non-maximum suppression to eliminate duplicate detections and retain the most accurate results.

In this project, a custom-trained YOLO model (e.g., *seafish.pt*) is used to specifically detect fish and marine objects. The model is loaded in the Python environment using the Ultralytics YOLO framework. Each frame from the video stream is processed, and detected objects are labeled with bounding boxes and class names using OpenCV.

The detection results are continuously updated and displayed on the web dashboard, allowing users to monitor fish presence and movement in real time. This helps in understanding fish behavior, detecting anomalies, and improving aquaculture management practices.

YOLO is highly efficient compared to other deep learning models, making it suitable for applications requiring low latency and real-time processing. Its integration into this system enhances the intelligence and automation of the monitoring process.

3.11 Code Description

The software implementation of the proposed system consists of two main parts: the embedded code running on the ESP32-S3 CAM module and the Python-based server code using Flask. These components work together to acquire data, process it, and display it through a web interface.

---

1. ESP32-S3 Embedded Code

The embedded code is written using the Arduino framework and is responsible for sensor interfacing, data acquisition, camera handling, and wireless communication.

Sensor Data Acquisition

The ESP32 reads data from multiple sensors:

- DS18B20 for water temperature
- DHT11 for surface temperature
- Ultrasonic sensor for water level
- BMP180 for pressure and altitude
- Turbidity sensor for water quality

Each sensor is initialized in the setup function, and readings are continuously updated in the loop function. The ultrasonic sensor calculates distance using the time-of-flight principle, while the DS18B20 and DHT11 provide digital temperature readings.

Camera Initialization and Capture

The ESP32-S3 CAM initializes the OV3660 camera using predefined GPIO configurations. The camera captures image frames and encodes them in JPEG format for efficient transmission. The capture handler is responsible for sending image frames when requested by the client.

Web Server Implementation

An HTTP server is created on the ESP32 using the ESP-IDF HTTP server library. It defines multiple endpoints:

- `/` → serves the web interface
- `/capture` → sends camera images
- `/sensors` → returns sensor data in JSON format

This allows external systems to access real-time data easily.

Wi-Fi Communication

The ESP32 connects to a Wi-Fi network using predefined SSID and password. Once connected, it acts as a server and shares its IP address for communication with the Flask application.

2. Python Flask Server Code

The Python backend is implemented using the Flask framework and is responsible for processing video streams, performing object detection, and serving the dashboard.

Video Stream Handling

The Flask server connects to the ESP32 video stream using HTTP requests. The MJPEG stream is decoded into individual frames using OpenCV. These frames are then processed in real time.

Object Detection using YOLO

A pre-trained YOLO model (*seafish.pt*) is loaded using the Ultralytics library. Each video frame is passed through the model to detect fish and other objects. Bounding boxes and labels are drawn on the frames using OpenCV functions.

Detected object labels are stored and updated continuously to provide real-time detection results.

Sensor Data Fetching

The Flask server periodically fetches sensor data from the ESP32 using HTTP requests. The data is received in JSON format and stored in a global dictionary for easy access and synchronization with detection results.

Multithreading

A separate thread is used to continuously fetch sensor data without interrupting the video processing stream. This ensures smooth and efficient operation of the system.

API Endpoints

The Flask application defines several routes:

- `/video` → streams processed video with detection overlays
- `/api/data` → returns combined sensor and detection data
- `/` → serves the web dashboard

3. Frontend Code (Web Dashboard)

The frontend is developed using HTML, CSS, and JavaScript, and is rendered using Flask.

- Displays live video stream with detection overlays
- Shows real-time sensor data
- Uses JavaScript to fetch updates asynchronously
- Provides system status indicators and visualization

The dashboard updates dynamically without reloading, ensuring a smooth user experience.

1. Wi-Fi Connection Setup (ESP32)

```
const char* ssid = "bps_wifi";
const char* password = "sagabps@235";

WiFi.begin(ssid, password);
while (WiFi.status() != WL_CONNECTED) {
    delay(400);
    Serial.print(".");
}
Serial.println("Connected → " + WiFi.localIP().toString());
```

Explanation

This code is used to connect the ESP32-S3 module to a Wi-Fi network. The SSID and password are provided, and the system continuously checks the connection status until it is successfully connected. Once connected, the ESP32 prints its IP address, which is used to access the web server and sensor data remotely.

---

2. Ultrasonic Sensor Distance Calculation

```
float getDistanceCM(){
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  duration = pulseIn(ECHO_PIN, HIGH);
  distance = (duration * 0.0343) / 2;
  return distance;
}
```

Explanation

This function measures the distance using the ultrasonic sensor. A trigger pulse is sent, and the time taken for the echo signal to return is measured. Using the speed of sound, the distance is calculated. This value represents the water level in the tank or pond.

---

3. Sensor Data JSON Response

```
snprintf(json, sizeof(json),
 "{\"distance\":%.1f,\"waterTemp\":%.1f,\"surfaceTemp\":%.1f,\"pressure\":%.1f,\"altitude\":%.1f,\"turbidity\":%d,\"quality\":\"%s\"}",
 currentDistance, waterTemp, surfaceTemp, pressure_hPa, altitude_m, turbidityValue, waterQuality.c_str());
```

Explanation

This code formats all sensor readings into a JSON string. JSON is used because it is lightweight and easy to transmit over HTTP. This data is later fetched by the Flask server and displayed on the dashboard.

---

4. Camera Frame Capture (ESP32)

```
camera_fb_t *fb = esp_camera_fb_get();
httpd_resp_send(req, (const char*)fb->buf, fb->len);
esp_camera_fb_return(fb);
```

Explanation

This snippet captures an image frame from the ESP32 camera and sends it as a JPEG response to the client. After sending the frame, memory is released. This enables real-time image streaming.

---

5. Flask Video Stream Handling

```
stream = requests.get(STREAM_URL, stream=True, timeout=10)

bytes_data = b''
```

Explanation

This code connects to the ESP32 video stream using HTTP. The stream is received as raw byte data, which is later decoded into image frames.

---

6. Frame Decoding using OpenCV

```
frame = cv2.imdecode(np.frombuffer(jpg, dtype=np.uint8), cv2.IMREAD_COLOR)
```

Explanation

This line converts raw JPEG byte data into an image frame that can be processed using OpenCV. It is a crucial step before applying object detection.

---

7. YOLO Object Detection

```
results = model(frame, verbose=False)
```

Explanation

This line passes the image frame into the YOLO model. The model processes the image and returns detected objects such as fish along with their bounding boxes and confidence scores.

---

8. Drawing Detection Boxes

```
cv2.rectangle(frame, (x1,y1), (x2,y2), (0,255,0), 2)

cv2.putText(frame, label, (x1, y1-10),

            cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)
```

Explanation

This code draws a rectangle around detected objects and displays their labels on the frame. This helps users visually identify fish in the video stream.

---

9. Sensor Data Fetching Thread

```
threading.Thread(target=sensor_poller, daemon=True).start()
```

Explanation

This creates a separate thread to continuously fetch sensor data from the ESP32. Using multithreading ensures that sensor updates do not interrupt video processing.

---

10. API Data Endpoint (Flask)

```
@app.route('/api/data')
def data():
    return jsonify({
        **sensor_data,
        "detections": latest_detections
    })
```

Explanation

This API endpoint sends combined sensor data and detection results in JSON format to the frontend. It allows real-time updates on the dashboard.

---

11. Live Dashboard Update (JavaScript)

```
let r = await fetch('/api/data');
let d = await r.json();
```

Explanation

This JavaScript code fetches real-time data from the Flask server and updates the dashboard dynamically without reloading the page.

Circuit Diagram

The circuit diagram represents the hardware connections of the smart aquaculture monitoring system centered around the ESP32-S3 CAM module. The ESP32-S3 acts as the main controller, interfacing with multiple sensors to collect environmental data and transmit it for processing.

The DS18B20 temperature sensor is connected using the One-Wire protocol. It is interfaced with a GPIO pin of the ESP32, along with a 4.7 kΩ pull-up resistor connected between the data line and VCC to ensure proper communication. This sensor is used to measure the water temperature.

The DHT11 sensor is connected to another GPIO pin of the ESP32 for measuring ambient temperature and humidity. It operates using a single digital data line along with power and ground connections.

The ultrasonic sensor (JNS-SR04T) is connected using two GPIO pins: one for the trigger signal and one for the echo signal. The ESP32 sends a trigger pulse and receives the echo signal to calculate the distance, which is used to determine the water level.

The turbidity sensor module is connected to the ESP32 through a digital input pin. It is powered using VCC and GND, and its output pin provides a signal indicating the water quality based on turbidity levels.

The BMP180 pressure sensor is interfaced using the I2C communication protocol. The SDA (data) and SCL (clock) lines are connected to the respective I2C pins of the ESP32-S3, enabling communication for pressure and altitude measurements.

All components are powered through a regulated 3.3V supply provided by the ESP32 or an external power module. Proper grounding is maintained across all components to ensure stable operation.

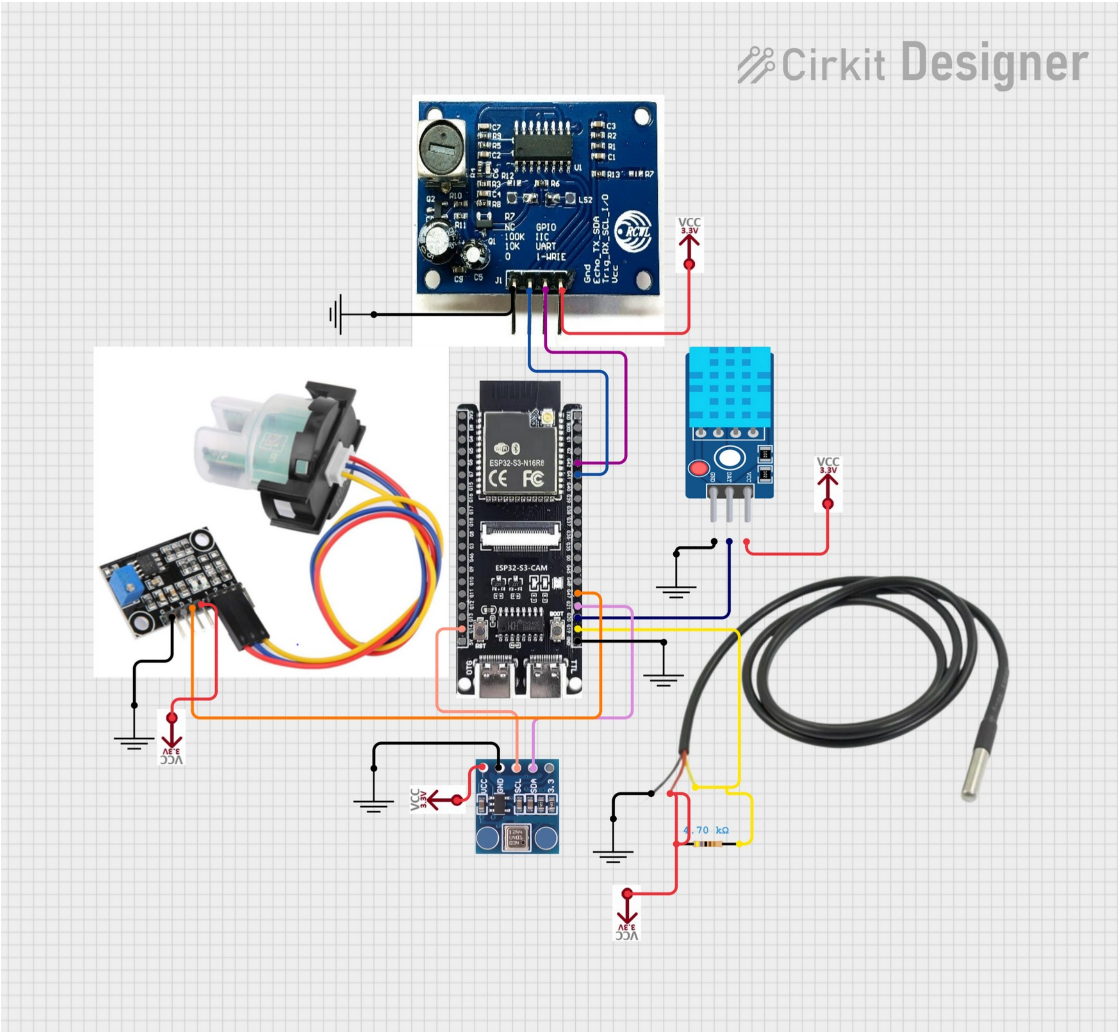

Fig-3.3: Circuit Diagram

The ESP32-S3 CAM module also includes a camera interface, which captures live video of the aquatic environment. The collected sensor data and video stream are transmitted via Wi-Fi to a local server for processing and visualization.

4. Result

The proposed smart fish detection and water monitoring system was successfully designed and implemented using the ESP32-S3 CAM module integrated with multiple sensors and an AI-based detection framework. The system was tested under real-time conditions to evaluate its performance in monitoring environmental parameters and detecting fish activity.

The ESP32-S3 module effectively collected data from all connected sensors, including water temperature (DS18B20), surface temperature (DHT11), water level (ultrasonic sensor), turbidity (water quality sensor), and atmospheric pressure (BMP180). The sensor readings were found to be stable and consistent, with minimal delay in data acquisition. The turbidity sensor successfully classified water quality into “Good” and “Bad” conditions, enabling quick assessment of the aquatic environment.

The camera module provided continuous image capture, and the video stream was transmitted successfully over a Wi-Fi network. The Flask-based server efficiently received the video stream and sensor data without significant packet loss. The integration of OpenCV allowed smooth frame decoding and processing.

The YOLO-based object detection model demonstrated effective performance in identifying fish within the video frames. The system was able to detect fish in real time and display bounding boxes with labels on the web dashboard. Detection accuracy was satisfactory under proper lighting conditions, although minor variations were observed in low-light or highly turbid environments.

The web-based dashboard displayed real-time data, including sensor values, detection results, and system status. The interface was responsive and updated dynamically without requiring manual refresh. Users were able to monitor both environmental parameters and fish activity simultaneously through a single platform.

Overall, the system achieved its objective of providing an integrated, real-time monitoring solution for aquaculture. The combination of IoT sensors and AI-based vision enhanced the system’s capability to deliver accurate and timely information. However, performance may vary depending on environmental conditions such as lighting, network stability, and water clarity.

The results demonstrate that the proposed system is a reliable and cost-effective solution for modern aquaculture monitoring, with potential for further improvements and scalability.

5. CONCLUSION

The proposed smart fish detection and water monitoring system has been successfully designed and implemented by integrating embedded systems, IoT technologies, and artificial intelligence. The system utilizes the ESP32-S3 CAM module as the core controller to collect environmental data from multiple sensors and to capture real-time video of the aquatic environment. The integration of sensors such as DS18B20, DHT11, ultrasonic sensor, turbidity sensor, and BMP180 enables continuous monitoring of critical water quality and environmental parameters.

The collected data is transmitted over a Wi-Fi network to a Flask-based server, where it is processed and displayed through a web-based dashboard. The use of OpenCV and a YOLO-based deep learning model allows real-time detection and identification of fish, enhancing the system's capability beyond traditional monitoring solutions. This combination of sensor data and visual intelligence provides a comprehensive understanding of aquaculture conditions.

The system demonstrated reliable performance in real-time operation, with accurate sensor readings and effective fish detection under suitable conditions. The web interface provided a user-friendly platform for monitoring both environmental parameters and fish activity simultaneously. The implementation of multithreading and efficient communication protocols ensured smooth data flow and minimal latency.

One of the major advantages of the proposed system is its cost-effectiveness and scalability. By using readily available components and open-source software, the system can be easily adapted and expanded for different aquaculture environments. It reduces manual effort, improves monitoring accuracy, and enables timely decision-making, which are essential for sustainable fish farming.

However, certain limitations were observed, such as reduced detection accuracy in low-light or highly turbid conditions and dependency on stable network connectivity. These challenges indicate potential areas for further improvement.

In conclusion, the developed system provides an efficient, intelligent, and integrated solution for modern aquaculture monitoring. It demonstrates the potential of combining IoT and artificial intelligence technologies to enhance productivity, sustainability, and environmental management in aquatic systems.